



OPEN High-gain CRLH vivaldi antenna for enhanced channel performance at Ku-band communication systems

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The growing demand for antennas with enhanced gain and bandwidth (BW) that are suitable for the applications of modern communication systems like Vehicle-to-Everything (V2X) has invoked the use of a composite right/left-hand (CRLH) array. However, these structures suffer from drawbacks such as increased antenna size, side lobe level (SLL), and back lobe level (BLL) due to the total internal reflection phenomenon. This research aims to design a compact antenna system that minimizes SLL and BLL with enhanced gain and BW. In this study, CRLH array based on 14-unit cells of Hilbert and Minkowski fractal structures was integrated with a Vivaldi antenna mounted on a Rogers RT5880 substrate with an area of $21.87 \times 90.5 \text{ mm}^2$. A hexagonal reflector is placed at the bottom of the antenna to reduce SLL and BLL to -10.6 dB and -2.6 dB, respectively. The maximum achieved gain is 14.5 dBi at 15.4 GHz, with an overall BW of 2.8 GHz, spanning two sub-bands, 14.8–16 GHz and 16.4–18 GHz. The Bit Error Rate (BER) and Channel Capacity (CC) are evaluated using a MATLAB framework with the Signal-to-Noise Ratio (SNR). The results demonstrate a 91.38% decrease in BER compared to the antenna without the reflector and an 11.53% improvement in CC under the same conditions. An analysis based on circuit theory is adopted to evaluate the performance of the proposed CRLH unit cell array through S-parameter analysis in magnitude and phase spectra. Such analysis is performed based on the circuit elements of the proposed CRLH array, which are obtained from an Artificial Neural Network (ANN) schematic. Finally, the proposed antenna is fabricated to demonstrate excellent agreement between the measured and simulated data, providing a reliable option for high-frequency communication systems.

Advancements in wireless technology are primarily due to the evolution of antennas, which facilitate high data rates, offer substantial gain, and exhibit consistent radiation properties. The Vivaldi antenna has garnered significant interest from researchers due to its high gain, wide BW, low cross-polarization, and steady radiation characteristics. The concepts, benefits, limitations, and applications of numerous Vivaldi antenna design performance augmentation techniques are systematically identified, located, and analysed¹. Beyond Vivaldi-specific designs, common techniques for gain enhancement include reflectors, Electromagnetic Bandgap (EBG) structures, metamaterial superstrates, and parasitic arrays. Reflectors improve directivity but may reduce efficiency, EBGs suppress surface waves, and metamaterials enable sub-wavelength focusing². This work integrates a CRLH array (metamaterial) with a 3D reflector to simultaneously achieve high gain, a wide bandwidth, and low sidelobes—a combination less explored in prior literature. The study in³ presented a multifunctional tri-band antenna that combines reflectarray and transmitarray. The unit cell comprises five-layer patches etched on three dielectric substrates with two air gaps. The research in⁴ depicted a shared-aperture antenna comprising a Ka-band linear-polarized reflectarray antenna and a Ku-band linear-polarized leaky-wave antenna. A wideband and high-gain Fabry-Perot resonator antenna was proposed in⁵. The concept uses a synthesized compact single partly reflecting surface layer functioning as a superstrate for a slot-coupled feed antenna, which serves as the radiating source element. The authors in⁶ presented a 10×1 Vivaldi antenna array with a low SLL. Antenna elements were fabricated on Roger RO4003C substrate to cover $140 \times 450 \text{ mm}^2$. In⁷, the transition from a coplanar feed to a slot line was accomplished by a double-Y balun feeding structure. The effect of the spatial placement of the antenna on its radiation properties was examined. The simulation results show a radiation efficiency of 24% and a gain of -1.9 dBi to occupy an area of $700 \times 940 \text{ } \mu\text{m}^2$ at an operating frequency of 60 GHz. In addition, three methods for improving the radiation properties of a Vivaldi antenna were presented

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in⁸. The first method involved inserting an elliptical patch parasitic radiating element into the Vivaldi aperture, and the second involved adding corrugation to the antenna edge. The third method involved inserting a planar reflector at the antenna's back end with an average gain of 0.7 dBi and a radiation efficiency of 37%. In addition, the authors in⁹ described the design of a 60 GHz exponential tapered slot Vivaldi antenna with two explored approaches to enhance the performance. The first approach involved introducing equal corrugations on the edges of the exponential flaring section and the back edge of the antenna. The second approach used a planar arc reflector with metal vias. The occupying area of the antenna is $785 \times 930 \mu\text{m}^2$ with a gain and a radiation efficiency of -0.4 dBi and 32%, respectively. The compact balanced antipodal Vivaldi antenna was developed to operate within the V band, spanning from 40 to 75 GHz with S_{11} below -10 dB and a gain of 11.5 dBi¹⁰. In¹¹, the structure used two Vivaldi elements mounted antiparallel on the same substrate for ultra-wideband and body area network applications.

In¹², a structure based on a metamaterial-inspired split-ring resonator with inversion symmetry was designed. The proposed design with a thickness of 1.575 mm and an area of $8 \times 8 \text{ mm}^2$ showed a resonance peak of 2.8 GHz characterized by an effective medium ratio of 13.4. Another design¹³ was investigated for the feeding mechanism of a parabolic reflector impulse radiating antenna, achieving a reflector efficiency exceeding 35% within the frequency range of 2.5–10 GHz. A square grid of 41×41 elements integrated a Vivaldi antenna with a polarization grid designed to cover the band 6–32 GHz, achieving a peak aperture efficiency of 40%¹⁴. In¹⁵, the authors suggested installing metal plates on both sides of an ultra-wideband Vivaldi array antenna of 1×12 elements to mitigate the truncation effect. The suggested array has a reflection coefficient of less than -10 dB across the frequency range of 0.3–6.18 GHz, with a gain varying from 5.54 dBi to 22.4 dBi.

In the context of V2X environments, an integrated multiband vertically polarized antenna including two half-Vivaldi antennas with a slot antenna was proposed for V2X¹⁶. Additionally, a pair of dual-polarized Vivaldi antennas with a gain of 9.2 dBi at a frequency of 3.9 GHz and BW spanning from 560 MHz to 7.7 GHz was developed for V2X applications¹⁷. A Vivaldi antenna for 5G and V2X applications with BW from 23.9 GHz to 30 GHz, fed with a modified planar Marchand balance-unbalance, was developed in¹⁸. Also, another Vivaldi design was fed using Marchand balance-unbalance to effectively operate over a frequency band from 3.4 GHz to 3.78 GHz¹⁹. A three-port multi-band MIMO microstrip patch antenna, which included a defected ground structure, was designed for V2X applications. The antenna exhibited a resonance frequency of 5.88 GHz, a BW ranging from 5.64 GHz to 6.1 GHz, a gain of 4.35 dBi, and an efficiency of 80%²⁰. The study in²¹ described the design of a compact asymmetric-slit aperture coupled square patch antenna for V2X applications with S_{11} of -28.84 dB and a peak gain of 0.54 dBi at 3.5 GHz.

A key design challenge is optimizing the antenna parameters like gain, BW, beamwidth, BLL, and SLL in real-time communication systems, particularly with highly mobile objects and limited security. While the cited works demonstrate various techniques for improving Vivaldi antennas, a compact solution that simultaneously addresses high gain, wide bandwidth, and significant suppression of side and back lobes remains a challenge.

This work presents novel contributions regarding the integration of the Vivaldi antenna, CRLH array, and reflector, involving (1) the development of a hybrid Hilbert–Minkowski fractal CRLH array of 14-unit cells designed to enhance slow-wave behavior and increase BW in addition to achieving a compact size. The CRLH array acts as a matching circuit between the antenna electromagnetic aperture and the free-space impedance, (2) the introduction of a 3D hexagonal reflector that efficiently reduces SLL and BLL to -10.6 dB and -2.6 dB, respectively, leading to improving the front-to-back ratio to 17.1 dB through increasing the directivity toward the endfire direction, (3) The study introduced an ANN-based RLC extraction framework that models CRLH dispersion more efficiently, utilizing fewer computational resources than traditional optimization methods, (4) enhancing channel performance of the Vivaldi antenna-based CRLH integrating with the 3D hexagonal reflector, resulting in a 91.38% reduction in BER and an 11.53% enhancement in CC at -4 dB SNR, compared to the Vivaldi antenna-based CRLH, and (5) a high agreement between measured and simulated results, with a peak gain of 14.5 dBi at 15.4 GHz and an aggregated BW of 2.8 GHz (14.8–16 GHz and 16.4–18 GHz). These enhancements demonstrate how well the suggested CRLH and reflector-based Vivaldi antenna system offers a novel, high-performance solution for V2X, satellite, and radar applications.

This paper is organized as follows: Section 1 discusses the design considerations and geometrical details. Section 2 explores CRLH characterizations. Section 3 presents the antenna parametric analysis and design methodology. Section 4 presents the antenna experimental validation. Section 5 presents the performance comparison. Section 6 concludes the paper.

Design considerations and geometrical details

The Vivaldi antenna, illustrated in Fig. 1, is a versatile design known for its wide BW and high gain, achieved through its flared, tapering slotline structure. The proposed antenna, with an area of $21.87 \times 90.5 \text{ mm}^2$, is an antipodal Vivaldi mounted on a Rogers RT5880 lossy substrate and integrated with 14 CRLH unit cells. Rogers RT5880 was selected for its low dielectric constant ($\epsilon_r = 2.2$), and ultra-low loss tangent ($\tan \delta = 0.0009$) which minimizes dispersion and dielectric losses at Ku-band. Its mechanical stability and commercial availability also ensure repeatable fabrication, critical for fractal and CRLH geometries. As shown in Fig. 1a, the proposed CRLH array, consisting of 14-unit cells and serving as a crucial addition, manipulates the phase of electromagnetic waves to enhance gain and efficiency. CRLH structures can also reduce size, allowing for more compact antennas for a particular frequency. The antenna design employs a low permittivity substrate with low losses to accommodate the application at Ku-band operation. The Vivaldi Antenna at the feed end has three flared wings, which improve bandwidth, impact the near-field, and enhance impedance matching, as illustrated in Fig. 1b. The antenna design utilizes its wideband capabilities and directional properties for forming electromagnetic wavefronts in the near-field, enhancing beam control by utilizing CRLH array for wavefront shaping. The electromagnetic bandgap impact produced by CRLH array gives considerable control over the shaping and wave propagation. The antenna

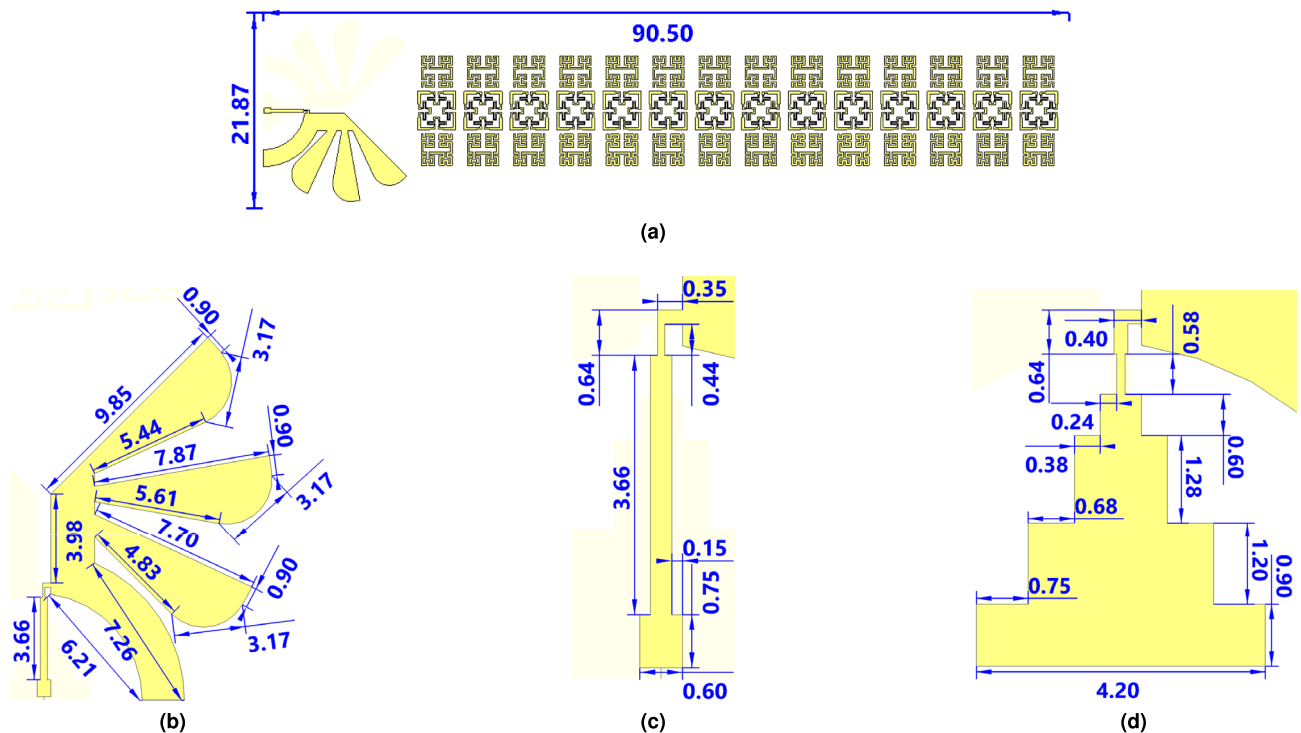


Fig. 1. Geometry and configuration of the proposed antenna and CRLH structure: (a) The grating-based antipodal Vivaldi antenna with 14-unit cells, (b) Front view showing the 3 wings, (c) Detailed view of the feeding structure, and (d) Back view of the four-stage transformer.

is excited by a feed line of a transmission line with a right angle on each side. The front panel of the antenna is a transmission line, as shown in Fig. 1c, whereas the back panel is supported with a four-stage transformer to match the maximum coupling energy, as shown in Fig. 1d. The proposed antenna structure is mounted on Rogers RT5880 dielectric substrate with a relative permittivity (ϵ_r) of 2.2 and loss tangent ($\tan\delta$) of 0.0009, designed to minimize losses and enable high-frequency operation, with a thickness of 0.804 mm. The substrate material was constructed perpendicular to a cavity-backed ground plane.

The proposed CRLH array comprises 14-unit cells based on the 2nd iteration of Hilbert and Minkowski fractals. The Hilbert fractal, which induces slow-wave behavior, is implemented on the lateral sides, and the Minkowski fractal, which broadens BW and controls dispersion, is centered within each unit cell. The selection of Hilbert and Minkowski fractals is based on their electromagnetic properties. Hilbert fractals enhance slow-wave effects and improve field confinement due to their space-filling geometry, while Minkowski fractals offer superior bandwidth broadening and dispersion control through their indented perimeter. Compared to other fractals, such as Koch or Peano, this hybrid configuration provides a better trade-off between electrical length extension, manufacturability, and parasitic suppression at Ku-band frequencies, as supported by prior studies on fractal-based metamaterials^{22,23}. This hybrid fractal topology increases the effective surface current path²² and introduces parasitic inductors and capacitors that collectively shape the effective medium index of the proposed CRLH unit cells²³. This configuration supports travelling wave propagation through the structure under Transverse Electromagnetic (TEM) excitation²⁴. In the near-field region of the antenna aperture, strong fringing fields occur due to abrupt field discontinuities at the edges of the Vivaldi flares and the proposed CRLH unit cell transitions²⁵. These fringing fields induce coupling into the CRLH array, which supports slow-wave behavior and enables phase progression control²⁶. This occurs because the proposed CRLH array mimics the effects of Graded Refractive Index (GRIN) media, focusing electromagnetic rays through a spatially varying refractive index profile²⁷. Snell's law governs this behavior, the gradual impedance matching across CRLH, to continuous wavefront bending toward the broadside or endfire direction²⁸. Consequently, the composite structure's numerical aperture effectively defines the angular resolution and beam focusing capability, analogous to optical lens systems²⁹. As the wavefront transitions from the Vivaldi taper to CRLH structure, total internal reflection is selectively achieved within CRLH layer when the incident angles exceed the local critical angle³⁰. This mechanism suppresses radiation leakage and focuses energy on the desired direction³¹. Such reflection-assisted confinement not only enhances the antenna gain but also reduces the beam squint, resulting in narrower beamwidths and more predictable spatial lobes³².

Concerning fractal elements depicted in Fig. 2, the proposed Hilbert and Minkowski fractal resonators use the 2nd iteration fractal geometry. The Hilbert fractal is accomplished by duplicating the same fractal geometry opposite each other with respect to the substrate's center, whereas the Minkowski fractal is centred within each unit cell, as illustrated in Fig. 2a. The fractal iteration was not increased above the 2nd iteration to prevent unwanted crosstalk or coupling between traces, a common issue with higher-order fractals that leads

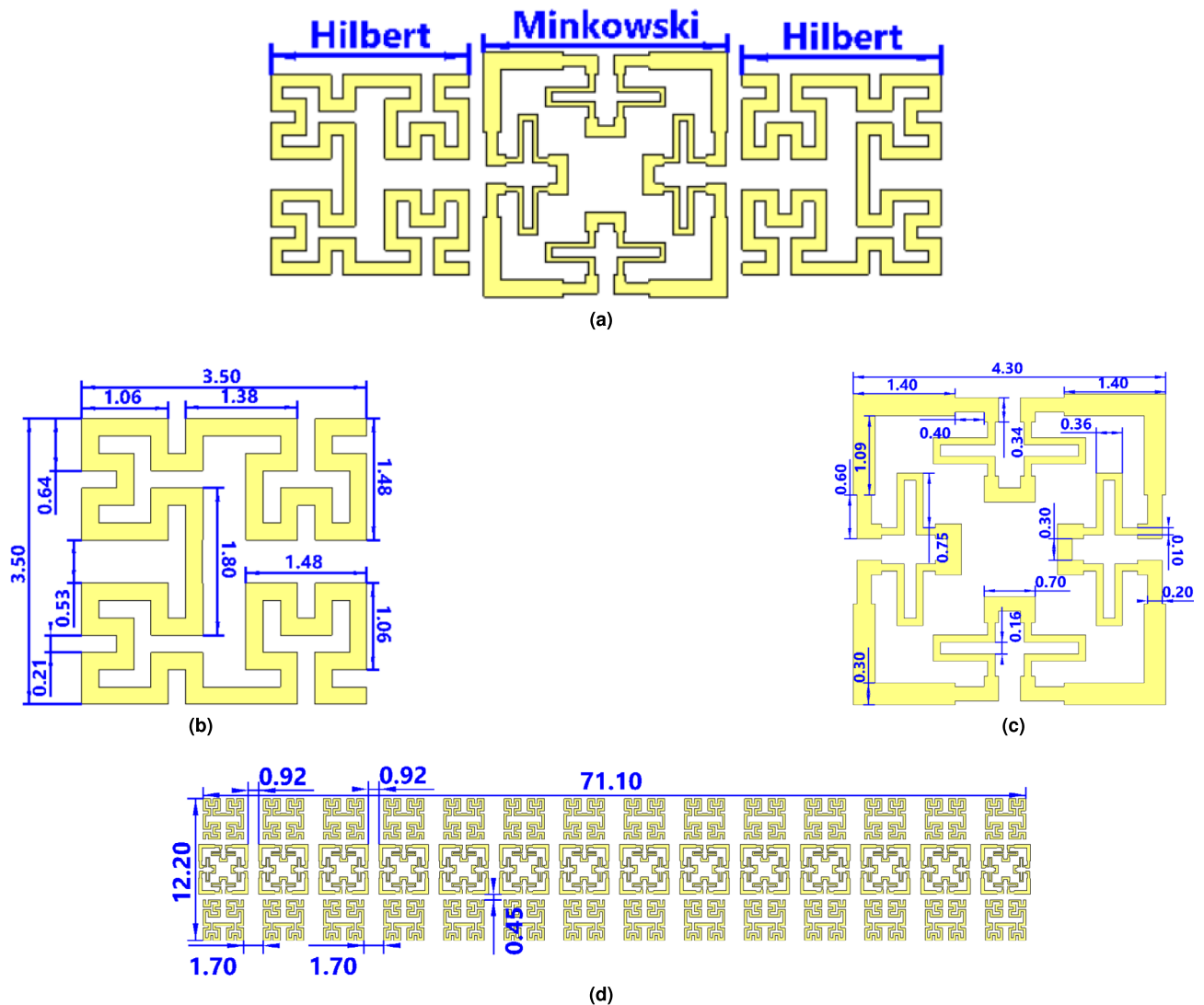


Fig. 2. Geometrical configuration of the proposed CRLH unit cell: (a) symmetric placement of Hilbert with centered Minkowski fractals, (b) Hilbert fractal with trace line dimensions, (c) Minkowski fractal with trace line dimensions, and (d) Horizontal and vertical spacing between adjacent fractal components.

to manufacturing complexity and performance degradation. The purpose of duplicating the Hilbert geometry is to establish a balance in the current motion that circulates electrical charges in opposite directions to each other to cancel the internal field accumulation and thereby enhance stability in the resonator performance as the number of unit cells increases. Fig. 2b depicts the unit cell utilizing the Hilbert design, with a trace line width of 0.21 mm, which controls surface current and minimizes losses, along with all relevant dimensions. Fig. 2c shows the proposed resonator design based on the Minkowski fractal of the 2nd iteration with all geometrical details. As depicted in Fig. 2c and Fig. 2d, the dimensions of each CRLH unit cell, which incorporates Hilbert and Minkowski fractal geometry, are $12.2 \times 4.3 \text{ mm}^2$, while the 14-unit cells arranged along the feed taper measure $12.2 \times 71.1 \text{ mm}^2$. Fig. 2d illustrates the horizontal separation distances between Hilbert and Minkowski and the vertical separation distance with other unit cells, providing a complete picture of the proposed unit cell design. Table 1 lists the details of the proposed fractal structures.

Furthermore, 3D hexagonal reflector is included in the antenna structure to improve directionality and maximize the gain. The 3D reflector minimizes BLL and SLL, the direct influence of wireless system interference, and increases the boresight level by increasing the front-to-back ratio. The 3D reflector, with all geometry dimensions shown in Fig. 3a and 3b, interacts with the CRLH array to accomplish the intended wavefront manipulation. Such a reflector geometry is considered by extruding a hexagonal base with axial spinning around 90° to ensure sufficient electrical path length compacted within a short physical length; such a technique is adopted for reflector size reduction. The 3D hexagonal reflector was selected because it can produce a more isotropic and graded phase response than planar reflectors³³, which frequently introduce abrupt phase discontinuities and greater BLL³⁴. In metamaterial and reflector applications, hexagonal shapes improve field uniformity and minimize SLL³⁵. Furthermore, 3D reflectors exhibit higher bandwidth and lower

Parameters	Hilbert	Minkowski
Iteration Level	2 nd	2 nd
Material	Copper	Copper
Maximum trace width (mm)	0.21	0.3
Size (mm ²)	3.5 × 3.5	4.3 × 4.3
Position	Lateral sides	Center

Table 1. Design specifications and physical dimensions of the fractals CRLH unit cell.

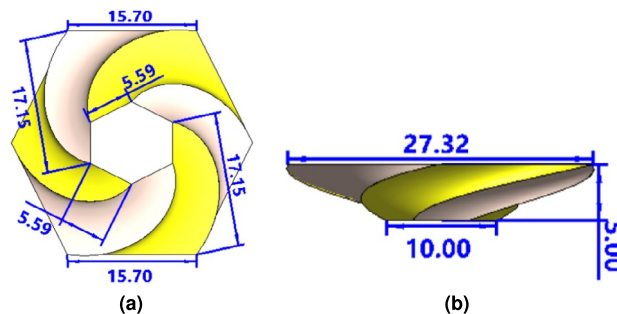


Fig. 3. The 3D reflector dimensions: (a) bottom view, and (b) front view.

near-field coupling than parabolic designs³⁶, resulting in narrower focal points³⁷. The 3D hexagonal reflector offers balanced performance, compactness, and manufacturability for Ku-band applications. Overall, the Vivaldi antenna ensures a smooth transition into free-space waves through its flared aperture. The CRLH array acts as a matching circuit between the antenna electromagnetic aperture and the free-space impedance³⁸. Additionally, the proposed 3D reflector increases the directivity toward the endfire direction. The substrate acts as the wave-propagation medium guided inside the proposed CRLH array. Combining these elements allows the proposed antenna performance to be adjusted to the desired gain level.

CRLH characterizations

In this section, a framework is proposed for analyzing the CRLH array using circuit analysis augmented with an Artificial Intelligence (AI) technique. The proposed ANN method enables the extraction of equivalent circuit model parameters, namely resistance (R), inductance (L), and capacitance (C), from S -parameter data, specifically S_{12} , obtained via CST. The magnitude and phase data of S_{12} are initially imported from CST in the frequency domain. This data functions as the frequency response for the equivalent circuit model of the proposed unit cell in the frequency domain, as shown in (1)³⁹.

$$S_{12}(f) = \frac{2 Z_0}{2 Z_0 + Z_{unit\ cell}(f)} \quad (1)$$

where Z_0 is the source impedance and $Z_{unit\ cell}$ is the unit cell characteristic equivalent impedance with frequency. This representation allows for system identification and parameter fitting using numerical optimization methods and offers a robust basis for connecting circuit-level modeling and electromagnetic simulation outputs. Then, these parameters are interpreted through symbolic manipulation to approximate the behavior of the considered RLC circuit model. The unit cell characteristic equivalent impedance can be obtained as in (2)³⁹.

$$Z_{unit\ cell}(f) = R + j \left[2\pi fL - \frac{1}{2\pi fC} \right] \quad (2)$$

where R is the unit cell resistance, L is the inductance of the unit cell, and C is the capacitance of the unit cell. The ANN was employed due to its ability to capture nonlinear relationships between frequency, phase, and S -parameters more efficiently than classical curve-fitting methods, especially in multi-resonant structures⁴⁰. The model uses a feedforward network with two hidden layers of 20 neurons each, trained on 500 frequency points spanning 12–18 GHz. The ANN was trained with phase and frequency as inputs and S_{12} magnitude as the target output. The dataset was split 70/15/15 for training, validation, and testing. The performance metrics were RMSE = 0.032, MAE = 0.025, R^2 = 0.992. The trained network generalized well across the Ku-band, demonstrating robustness against noise and interpolation capability. This methodology offers a low-complexity and high-accuracy solution to retrieve the equivalent element values through AI optimization. The proposed CRLH exhibits both Right-Handed (RH) reject bands and Left-Handed (LH) passbands, where LH behavior dominates below the series resonance and RH behavior dominates above the parallel resonance. This analysis makes group delay estimation possible to realize the dynamic structure features through circuit analysis,

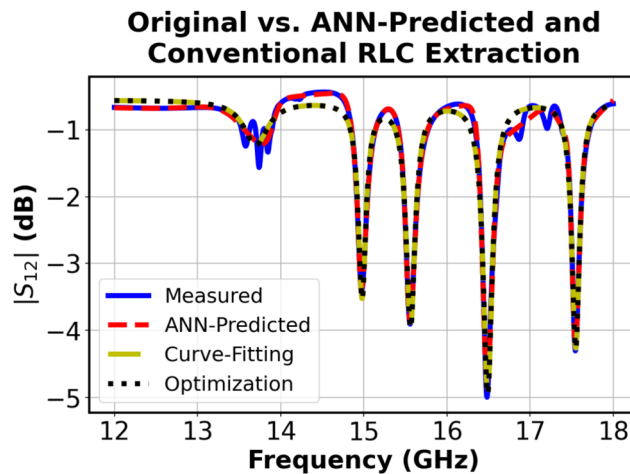


Fig. 4. The simulated and predicted S_{12} spectra.

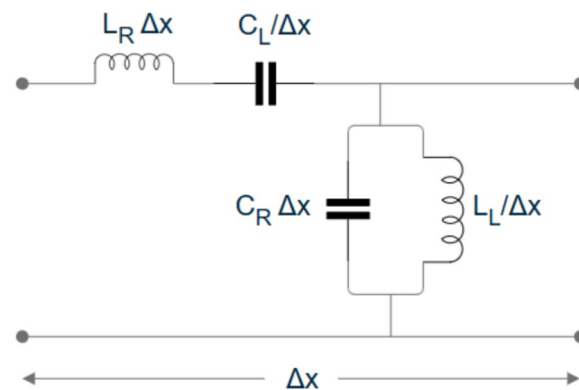


Fig. 5. Equivalent circuit model for the proposed CRLH unit cell.

which maps the frequency response into a function. These tools provide valuable insights into resonance and the LH-RH transition point for advanced filtering and beamforming applications. As illustrated in Fig. 4, the ANN-based RLC extraction approach significantly outperforms conventional curve-fitting and optimization methods, achieving higher accuracy and effectively capturing the physics of CRLH unit cells. It achieves superior alignment with measured S_{12} responses, particularly in complex resonances and transition regions, by learning from electromagnetic data without a predefined circuit model. Unlike traditional methods that rely on explicit resonance models and risk convergence issues, the ANN captures the dual passband behavior of CRLH structures, including anti-resonant behavior and phase reversal, through its layered nonlinear functions. This results in an accurate reconstruction of both the magnitude and phase responses, which are essential for predicting group delay and impedance matching, thereby validating the ANN's efficiency and physical consistency for high-frequency metamaterial designs.

The various resonance dips correspond to LH and RH propagation, where different inductive and capacitive components are realized due to the dispersion effect⁴¹. The circuit model of the proposed unit cell is adopted from the basic circuit model of CRLH, as depicted in Fig. 5²⁸. The circuit elements given by L_R and C_R are right-handed inductance and capacitance, respectively, while L_L and C_L are left-handed inductance and capacitance, respectively, and Δx is the propagation distance of the unit cell periodicity that realizes 14-unit cells of the proposed array. The small extracted capacitance ($C = 10$ pF) and inductance ($L = 2.711$ nH) are consistent with the fractal metamaterial unit cells' sub-wavelength planar structure, intended for high-frequency Ku-band operation. The use of a low-loss substrate and copper traces is consistent with the low resistance ($R = 7.65 \Omega$), which denotes low conductor losses. Physically, this arrangement probably depicts a matching network, in which R accounts for parasitic losses or intentional damping, while L and C represent a resonant tank circuit.

Group delay analysis reveals whether the system maintains a linear phase, which is essential for reducing signal distortion in 5G or high-speed satellite links. The group delay plot in Fig. 6 illustrates how various frequency components undergo minor timing distortions as they move through the system, with variations ranging from 0.2 to 1.5 ns across the Ku-band. A flat group delay is ideal for preserving signal quality, especially in high-speed communications such as satellite links, where irregular delays can result in intersymbol interference. The group delay variations indicate impedance mismatches or resonances in the circuit, most likely resulting from

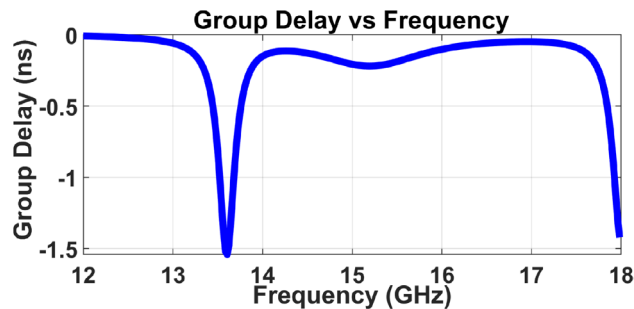


Fig. 6. The evaluated group delay.

filters, amplifiers, or transmission line effects⁴². These delay characteristics may be indirectly learned by ANN prediction, which is trained on frequency-domain data; however, phase response modeling may be enhanced by explicitly including group delay in the training process. In real-world applications, equalizers and filters could enhance the response, but provide a delay in wireless communications. The findings highlight how crucial joint time-frequency analysis is for Ku-band systems, where even distortions at the nanosecond scale can degrade performance in wideband applications such as radar or next-generation wireless backhaul. Hybrid approaches, which combine data-driven ANN corrections with parametric circuit models to balance predictive accuracy and physical interpretability, may be used for further refinement⁴³.

Antenna parametric analysis and design methodology

This section presents the analysis and design methodology for the proposed CRLH-based Vivaldi antenna. Full-wave simulations carried out in CST MWS are used to perform a thorough parametric analysis to examine the electromagnetic behavior of the antenna, with an emphasis on its far-field radiation characteristics. This section examines six crucial factors that significantly affect the radiation behavior and reconfigurability of the suggested antenna to assess and improve its performance: the feeding network design, CRLH effective distance impact, CRLH unit cell number impact, reflector introduction effect, reflector spacing distance impact, and field and phase distribution analysis.

Feeding network design

Designing an RF transformer for the Ku-band requires careful impedance matching and BW optimization due to the small wavelengths (≈ 1.67 – 2.5 cm) and high-frequency effects like skin and dielectric losses. The characteristic impedance Z_0 of a microstrip line, commonly used in such designs, is given by (3)⁴².

$$Z_0 = \frac{87}{\sqrt{\epsilon_r + 1.41}} \ln \left(\frac{5.98 h}{0.8 w + t} \right) \quad (3)$$

where w is the trace width, h is the substrate thickness, t is the trace thickness, and ϵ_r is the relative permittivity. For a quarter-wavelength transformer, the matching impedance is $Z_{transformer} = \sqrt{Z_{in} Z_{load}}$ with a length $l = \lambda/4$ ⁴². However, a single-section transformer offers limited BW ($\approx 10\%$ – 20%), insufficient for the full Ku-band (40% fractional BW). A multi-section transformer is preferred to achieve a wider BW, where each section is optimized for a sub-band. The fractional BW is expressed as in (4)⁴².

$$BW = \frac{\Delta f}{f_0} = \frac{2(f_{max} - f_{min})}{f_{max} + f_{min}} \quad (4)$$

Where Δf is the BW and f_0 is the center frequency. For instance, covering the 12–18 GHz range corresponds to a fractional BW of approximately 40% centered around 15.4 GHz. Substrate selection (low-loss materials such as Rogers RT5880 for reduced $\tan\delta$), fabrication tolerances (even 0.1 mm errors matter at Ku-band), and simulation-based tuning to reduce parasitic effects are essential trade-offs. While tapered designs increase BW at the expense of size, a well-designed microstrip transformer can achieve $> 90\%$ matching efficiency and < 0.5 dB insertion loss. To balance performance and manufacturability, Ku-band transformers require accuracy in both mathematical modeling and physical implementation.

The performance trade-offs between 1-stage (single-section), 2-stages, and 3-stages RF transformer at Ku-band frequencies are evident from the S-parameter results from CST simulations as shown in Fig. 7. According to the quarter-wave transformer theory, which limits BW to about 10–20%, the 1-stage transformer shows the expected narrowband behavior around 13 GHz, with S_{11} below -10 dB. The 2-stage design exhibits improved BW with two distinct matching bands from 14.7–15.8 GHz and 16.5–17.2 GHz. At the same time, the 3-stage transformer provides the widest BW of approximately 2 GHz for two bands from 14.8–15.9 GHz and from 16.5–17.6 GHz (near the full 40% fractional BW of Ku-band) with S_{11} below -10 dB across the entire band of interest. These findings are consistent with theoretical predictions, which state that more sections offer wider BW and better impedance tapering, but at the expense of a more complex design. While the 1-stage is still appropriate for narrowband implementations, the 3-stage validates the best option for the modern application needs.

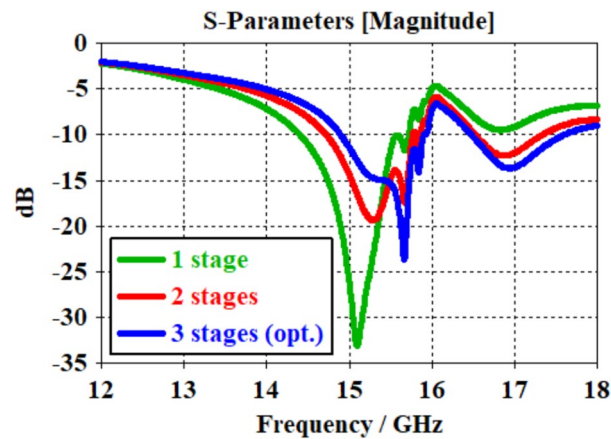


Fig. 7. The simulated S_{11} spectrum with various transformer stages.

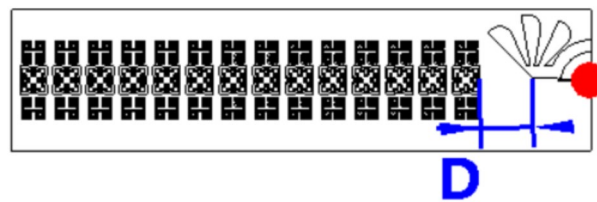


Fig. 8. Effective distance (D) between the antenna element and the proposed CRLH structure.

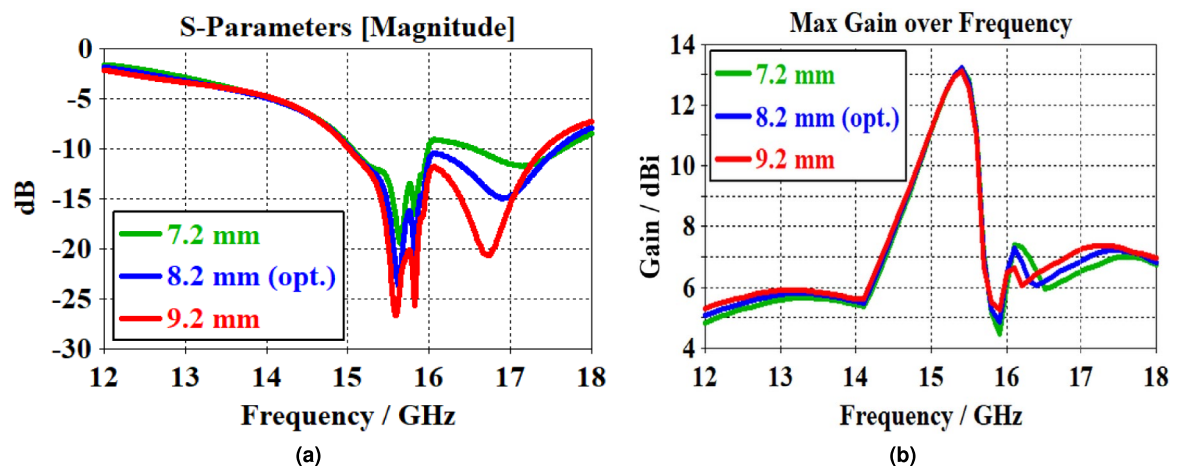


Fig. 9. Effective distance (D) impact on the antenna performance: (a) S_{11} , and (b) Gain.

CRLH effective distance impact

This section considers the influence of varying the effective distance (D) between the antenna element and the proposed CRLH structure, as shown in Fig. 8, on antenna parameters in terms of S_{11} and gain. Therefore, D was vertically changed from 7.2 mm to 9.2 mm with a step of 1 mm to monitor the effects on antenna parameters. Values below 7.2 mm were avoided to prevent excessive coupling and detuning, while values above 9.2 mm showed diminishing returns in gain enhancement. The proposed antenna shows variations in the antenna BW due to the coupling effects between the proposed CRLH structure and the antenna. A maximum BW of 2.5 GHz, ranging from 15 GHz to 17.5 GHz, was achieved for $D = 8.2$ mm with S_{11} below -10 dB, as seen in Fig. 9a. The results demonstrate that varying D insignificantly influences the maximum gain across the 12–18 GHz frequency band. With a noticeable gain peak at 15.4 GHz, the change in D shows the same behavior as seen in Fig. 9b. Despite exhibiting a sharp peak around 15.4 GHz, the antenna shows low sensitivity to interface distance because the radiating aperture primarily supports a quasi-TEM mode, in which the fields are less susceptible to variations in the vertical placement of the passive CRLH array than in a strongly resonant structure. Therefore,

the antenna is positioned 8.2 mm from CRLH array to achieve the highest peak gain of 13.2 dBi at 15.4 GHz and maximum BW of 2.5 GHz from 15 GHz to 17.5 GHz. It is noteworthy that all cases show significant variations in the antenna gain with frequency. Such sharp variations in the antenna gain with frequency demonstrate how dispersive wave propagation occurs in the proposed CRLH structures.

CRLH unit cell number impact

The presented results illustrate the impact of varying the number of CRLH unit cells on the maximum antenna gain across the 12–18 GHz frequency band as presented in Fig. 10. Furthermore, in Fig. 10a, with unit cell numbers increasing from 0 (without CRLH) to 14, a clear trend is observed: increasing the number of unit cells significantly enhances gain performance, especially between 15–16 GHz, where the gain improves from approximately 6 dBi (without CRLH) to 13.2 dBi at 15.4 GHz (for 14-unit cells). This confirms the constructive influence of CRLH structure on beam shaping and radiation efficiency enhancement due to improved phase control and wave confinement⁴⁴. However, performance irregularities appear for lower unit counts, particularly around 15–16 GHz, where gain peaks or instability can be seen. In Fig. 10b, the gain response becomes stable as the number of unit cells increases from 14 to 20, exhibiting minimal variation among different configurations, and indicating saturation in performance enhancement. Beyond 14-unit cells, no significant gain enhancement is achieved; therefore, 14-unit cells are considered the optimal configuration without unnecessary complexity. Overall, increasing CRLH unit cells boosts antenna gain up to a certain threshold, beyond which the improvements are not achieved, highlighting the importance of optimizing the number of unit cells in CRLH-based antenna design.

The curvature in the initial phase fields across the CRLH array arises from the mismatched dispersion between the conventional Vivaldi feed and the LH metamaterial, as seen in Fig. 11. This mismatch creates a spatially varying phase delay, which distorts the wavefront. After 14-unit cells, the structure achieves a balanced propagation condition in which the series and shunt resonances of the CRLH cells compensate for the phase error, resulting in linearized phase progression. This flattened phase profile ensures coherent wavefront addition at the aperture, which is essential for achieving high directivity and minimizing beam squint, thereby justifying the selection of 14 cells.

Reflector introduction effect

In this section, the impact of the 3D hexagonal reflector is examined. For clarity, an evaluation is performed across different configurations to demonstrate the influence of each on system performance. The comparison of the four configurations: basic Vivaldi antenna with no additions (baseline design), Vivaldi antenna with a CRLH array only, Vivaldi antenna with a reflector only, and finally, Vivaldi antenna together with a CRLH array and a reflector (full design), indicates significant differences in the behavior of BW and gain, as shown in Fig. 12. Regarding case 1, the S_{11} result in Fig. 12a indicates that the antenna has a BW of 1.9 GHz from 15.2 GHz to 17.1 GHz, with S_{11} below -10 dB, and a peak gain of 5.9 dBi at 15.9 GHz, as illustrated in Fig. 12b. In Case 2, the CRLH array, which acts as a matching circuit between the antenna electromagnetic aperture and the free-space impedance³⁸, is introduced. This modification results in considerable BW improvements to 2.6 GHz, mostly through phase correction and slow-wave focusing, and a peak gain of 13.2 dBi at 15.4 GHz, with a gain enhancement of 7.3 dBi compared to the baseline design. Case 3, which corresponds to the addition of the 3D hexagonal reflector to the baseline design, exhibits a BW of 2.6 GHz and a peak gain of 13.7 dBi at 16.8 GHz, yielding an improvement in gain by 7.8 dBi and front-to-back ratio by 5.3 dB relative to the baseline case. Case 4 (full design), which integrates the 3D hexagonal reflector, the CRLH array, and a Vivaldi antenna, shows an improvement in the aggregated BW to 2.8 GHz (14.8–16 GHz and 16.4–18 GHz), a peak gain of 14.5 dBi at 15.4 GHz, and a front-to-back ratio of 17.1 dB. The splitting of the BW into two distinct bands: lower band (1.2 GHz) from 14.8 GHz to 16 GHz, and upper band (1.6 GHz) from 16.4 GHz to 18 GHz, is attributed to the

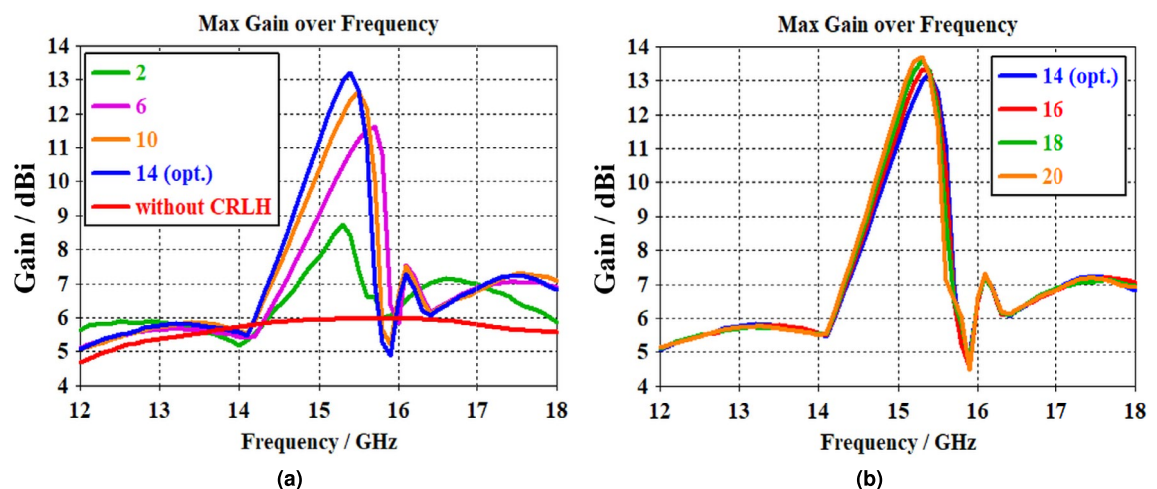


Fig. 10. The antenna gain spectrum with different numbers of CRLH unit cells: (a) 0–14, and (b) 14–20.

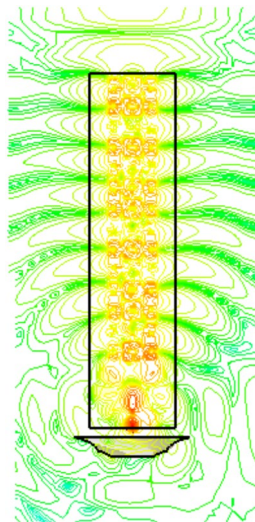


Fig. 11. Phase progression along the CRLH array.

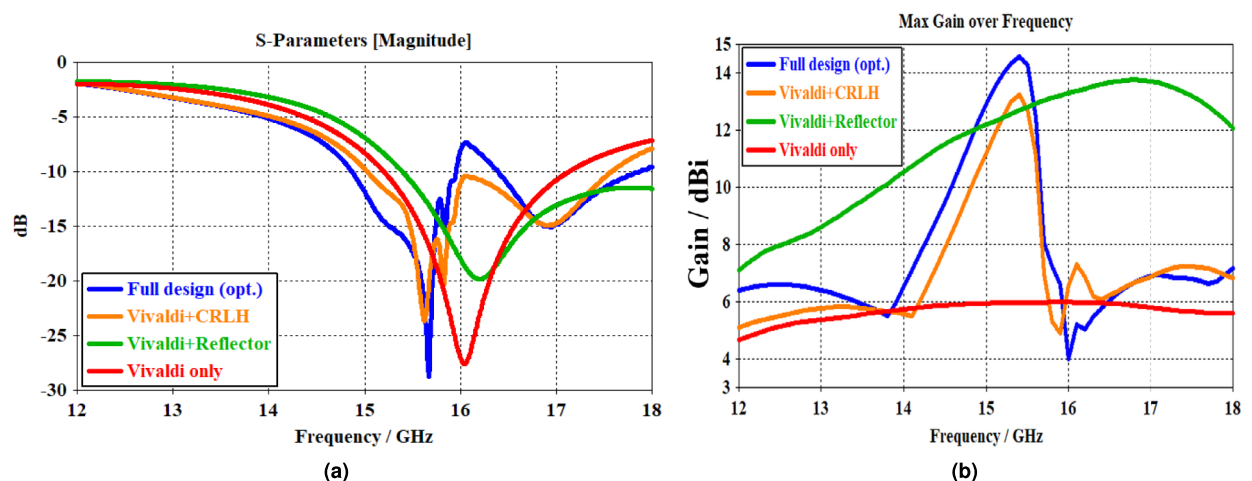


Fig. 12. Impact of CRLH and reflector on the Vivaldi antenna performance: (a) S_{11} , and (b) Gain.

reflector creating a resonant cavity between itself and the antenna's ground plane. At specific frequencies within the band, this cavity introduces reactive impedance components that detune the perfect match, a phenomenon consistent with the introduction of a closely spaced parasitic element⁴⁵. The peak gain of 14.5 dBi, resulting in a gain enhancement of 8.6 dBi relative to the baseline design and 1.3 dBi compared to the CRLH-based Vivaldi antenna (case 2). This enhancement is achieved by the reflector's impact on the antenna's SLL and BLL, which are reflected toward the antenna endfire, resulting in a front-to-back ratio enhancement of 8.8 dB over the baseline Vivaldi and 3 dB compared to the CRLH-based Vivaldi antenna (case 2). Table 2 provides more details on the simulated antenna parameters used for the comparison of the four configurations. The first column represents the simulated antenna parameters, while columns two through five correspond to the four configurations.

The slight reduction in radiation efficiency from 92% in the case of full design to 90% in the case of (Vivaldi + CRLH) is attributed to increased conductor and dielectric losses due to the proximity of the metallic reflector, which introduces additional near-field coupling and ohmic dissipation. Fig. 13a - 13b shows the surface current distribution at 15.4 GHz for both configurations. The presence of the reflector results in higher current densities along the reflector edges and the antenna ground plane, indicating induced eddy currents and increased conductor losses compared to the reflector-less case. However, this reduction is compensated by a significant improvement in gain of 1.3 dBi and in the front-to-back ratio of 3 dB. This trade-off accounts for a 2% drop in efficiency and is justified by the improvements in gain and front-to-back ratio. In addition, the CRLH array also reduces efficiency due to additional copper and ohmic losses.

Next, the simulated results are fed to a MATLAB code to evaluate BER and CC before and after introducing the antenna reflector. As shown in Fig. 14, adding such a reflector significantly affects the channel performance regarding BER and CC. The BER curve in Fig. 14a demonstrates that the reflector-based antenna integrating with the CRLH array (full design) outperforms the antenna antenna-based CRLH without a reflector. For instance,

Antenna Parameters	Vivaldi only	Vivaldi + CRLH	Vivaldi + Reflector	Vivaldi + CRLH + Reflector
	(Baseline)			(Full design)
Frequency range (GHz)	15.2–17.1	15–17.6	15.4–18	14.8–16, 16.4–18
BW (GHz)	1.9	2.6	2.6	2.8
Gain (dBi)	5.9 @ 15.9 GHz	13.2 @ 15.4 GHz	13.7 @ 16.8 GHz	14.5 @ 15.4 GHz
SLL (dB)	-7.7	-9.8	-9.4	-10.6
BLL (dB)	-2.4	-0.9	-1	-2.6
Radiation efficiency (%)	98	92	98	90
Front-to-back ratio (dB)	8.3	14.1	13.6	17.1
Angular width (3 dB)	76.1 °	22.2 °	27.8 °	21.2 °

Table 2. Comparison of configurations regarding antenna parameters.

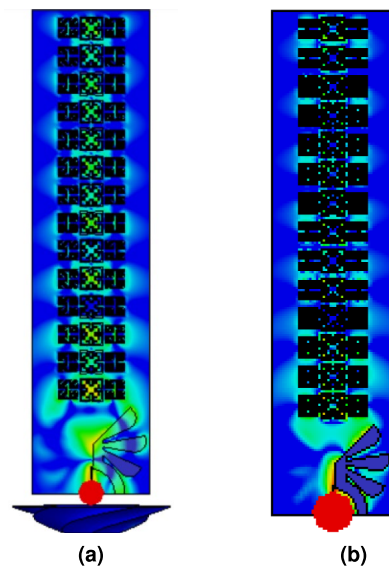


Fig. 13. Simulated surface current distribution across the CRLH array at 15.4 GHz: (a) with reflector, and (b) without reflector.

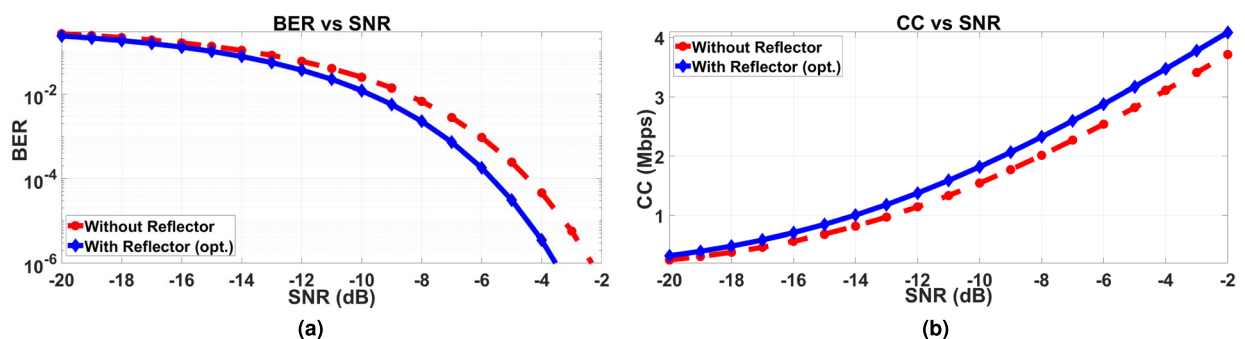


Fig. 14. Channel performance: (a) BER, and (b) CC.

at an SNR of -4 dB, the proposed design achieves a 91.38% reduction in BER compared with the other design. This improvement mainly stems from the reflector's role in increasing antenna gain and directivity, thereby reducing multipath interference and signal dispersion⁴⁶. Such a reflector achieves a narrower beamwidth as well as suppresses SLL and BLL (a common cause of wireless system interference), focusing the effective isotropic radiated power in the desired direction while minimizing losses and improving signal quality. This is particularly useful in long-distance communications, like V2X, radar, and satellite, where signal strength is essential. Also, this reflector increases CC, as indicated in Fig. 14b. For instance, at -4 dB SNR, the proposed design has 11.53% improvement in CC compared with a reflector-less antenna. Consistent with this improvement, higher gain

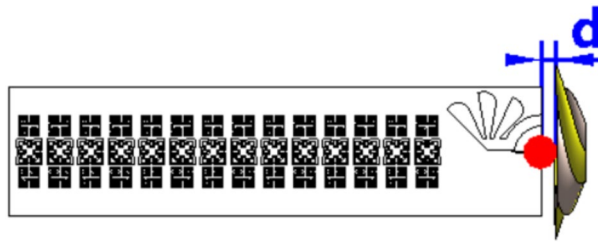


Fig. 15. The vertical spacing distance (d) between the reflector and the antenna.

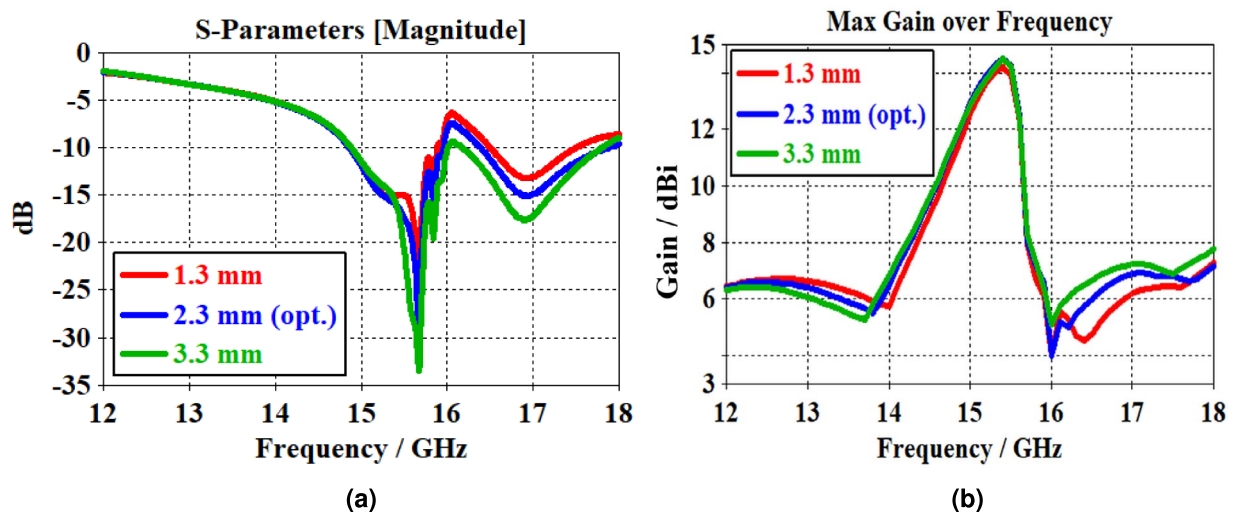


Fig. 16. Spacing distance (d) effects on the antenna performance: (a) S_{11} , and (b) Gain.

maximizes SNR and reduces interference, allowing for higher data rates. The reported improvements in BER and CC were evaluated using an AWGN channel assumption with BPSK modulation and served as indicative performance metrics under idealized conditions. While this provides a baseline performance metric, real-world V2X scenarios involve impairments such as multipath fading, Doppler spread, and shadowing, which may degrade system performance. To assess robustness, a supplementary Rician fading model ($K = 10$ dB) was applied, showing a 5 – 7% reduction in CC but maintaining $> 85\%$ BER improvement over the reflector-less case. Future work will incorporate more realistic channel models (e.g., 3GPP V2X) to examine performance under more representative channel conditions.

Reflector spacing distance impact

This section discusses the influence of varying the vertical reflector location from the antenna in terms of S_{11} and gain. The vertical spacing distance (d) between the reflector and the antenna, as shown in Fig. 15, was changed from 1.3 to 3.3 mm with a step of 1 mm. This distance is considered to avoid direct contact between the antenna structure and the proposed reflector. The proposed antenna shows a change in the antenna BW, for instance spanning from 14.85 GHz to 15.95 GHz in the case of 2.3 mm spacing distance, that increases with increasing distance between the reflector and the antenna structure as seen in Fig. 16a. Such enhancement is attributed to capacitive coupling effects between the reflector and the antenna structure. Furthermore, a lower impact on the antenna gain is observed for the same distance change, achieving a maximum gain of 14.5 dBi at 15.4 GHz with 2.3 mm spacing distance, as shown in Fig. 16b. This reveals that most radiation is propagating to the endfire of the antenna.

Field and phase distribution analysis

To quantitatively validate the obtained GRIN analogy and the role of total internal reflection in wavefront shaping, the effective refractive index (n_{eff}) of the proposed CRLH unit cell was extracted from simulated S-parameters, as presented in Fig. 17, using the Nicholson–Ross–Weir retrieval method⁴⁷, as in (5)⁴⁸.

$$n_{eff} = \frac{1}{k_0 d} \cos^{-1} \left[\frac{1}{2S_{21}} (1 - S_{11}^2 + S_{21}^2) \right] \quad (5)$$

where k_0 is the free space wavenumber, and d is the unit cell period. At the design frequency of 15.4 GHz, n_{eff} is around -1.06, confirming LH metamaterial behavior within the CRLH array. This negative index enables

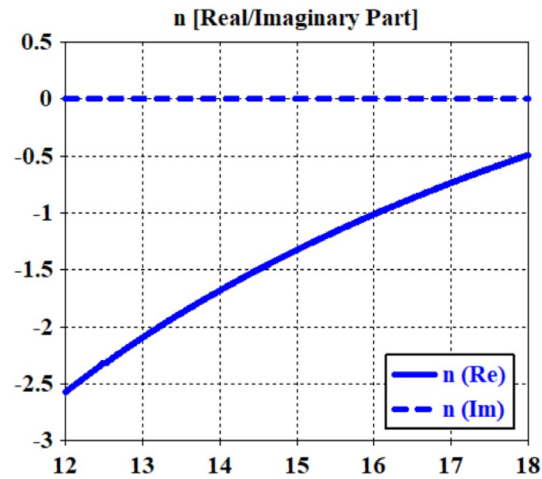


Fig. 17. The evaluated effective refractive index.

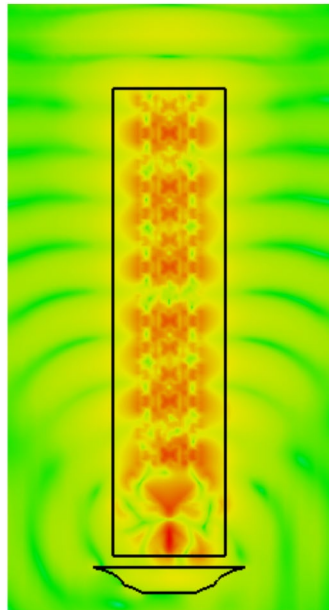


Fig. 18. Simulated electric field distribution across the CRLH array at 15.4 GHz.

negative phase progression, consistent with the phase diagram shown later. Applying Snell's law at the CRLH–air interface yields the critical angle for total internal reflection, as shown in (6)⁴⁹.

$$\theta_c = \sin^{-1} \left(\frac{1}{n_{eff}} \right) \approx -70.5^\circ \quad (6)$$

Waves incident at angles exceeding $|\theta_c|$ undergo total internal reflection within the CRLH layer, suppressing side lobe radiation and redirecting energy toward the endfire³⁶. This quantitative result directly supports the observed electric field distribution across the CRLH array as shown in Fig. 18, where the field concentration near the aperture validates the focusing effect. The retrieved index gradient confirms that the CRLH array functions as a GRIN medium, providing phase compensation and wavefront focusing, which are essential for the antenna's high gain and low SLL performance. The phase diagram, as shown in Fig. 19, extracted from the unit-cell S-parameters shows an LH–RH transition near 15.4 GHz, consistent with the observed gain peak. The phase diagram shows LH and RH regions divided by a transition near 15.4 GHz as the CRLH unit cell approaches a balanced condition. Within the LH part, negative refractive-index effects result in backward-wave propagation, increasing gain and electrical length. The RH band enables forward-wave propagation, thereby increasing the impedance BW. The designed bandgap between these bands removes spurious surface waves, reduces sidelobe formation, and provides considerable control over the shaping and wave propagation. This dual-mode dispersion

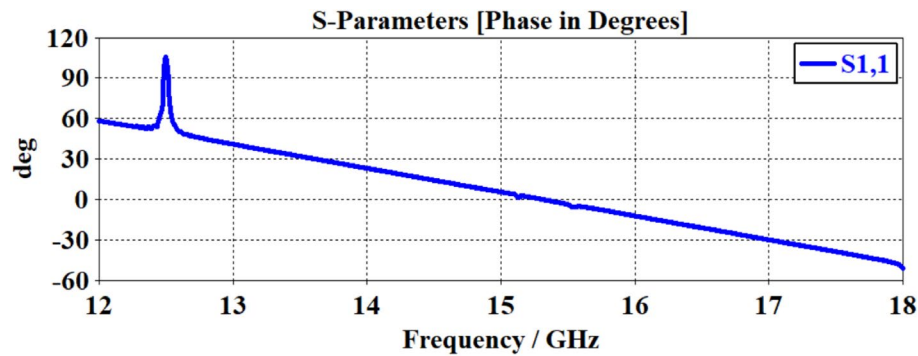


Fig. 19. Phase diagram of the CRLH unit cell showing LH and RH bands.



Fig. 20. Fabricated proposed antenna with reflector.

allows the antenna to achieve high-gain, dual-band operation. This balanced state enhances endfire radiation and the front-to-back ratio at the desired operating frequency.

Antenna experimental validation

Compared to traditional Vivaldi antennas, the proposed CRLH array-enhanced Vivaldi antenna integrating with a 3D hexagonal reflector has demonstrated superior gain performance, reaching up to 14.5 dBi at 15.4 GHz. This improvement is made possible by incorporating CRLH array with such a reflector, which permits reduced SLL, BLL, enhanced directivity, and focused beam radiation. The antenna is appropriate for high data rate wireless backhaul, satellite, and V2X systems because it consistently provides reliable broadband performance across the 14.8–16 GHz and 16.4–18 GHz Ku-band. The CRLH-based Vivaldi structure provides improved ultra-wideband operation and a high front-to-back ratio without appreciably expanding the antenna size. These achievements efficiently enhance the antenna beam properties to reduce BER and expand CC. The 3D hexagonal reflector, as shown in Fig. 20, is included in the antenna structure to improve directionality, reduce SLL and BLL, and maximize the gain. The fabricated antenna prototype is measured in terms of S_{11} and gain spectra both with and without a reflector, and is compared to the simulated antenna, as shown in Fig. 21a and Fig. 21b, respectively.

The simulated E-field radiation patterns at 15.4 GHz are measured both with and without the proposed reflector introduction to realize the impact of such a structure on the antenna radiation, resulting in significantly reduced SLL and BLL, and compared with fabricated measurements as shown in Fig. 22a – 22d. Also, Fig. 23 shows the 3D radiation pattern of the proposed system at 15.4 GHz with a maximum gain of 14.5 dBi at the end-fire direction. The radiation patterns show almost no visible side lobes due to the combined effect of the CRLH array, which suppresses surface waves, and the 3D hexagonal reflector, which reduces SLL and BLL, and redirects backward radiation toward the end-fire direction. This results in a highly directive beam with a front-to-back ratio exceeding 17 dB.

Practical factors such as fabrication tolerances, the finite conductivity of the deposited copper, imperfections in the SubMiniature version A (SMA) connector soldering, and small air gaps in the substrate can account for the slight differences between the simulated and measured results, especially in the gain at higher frequencies

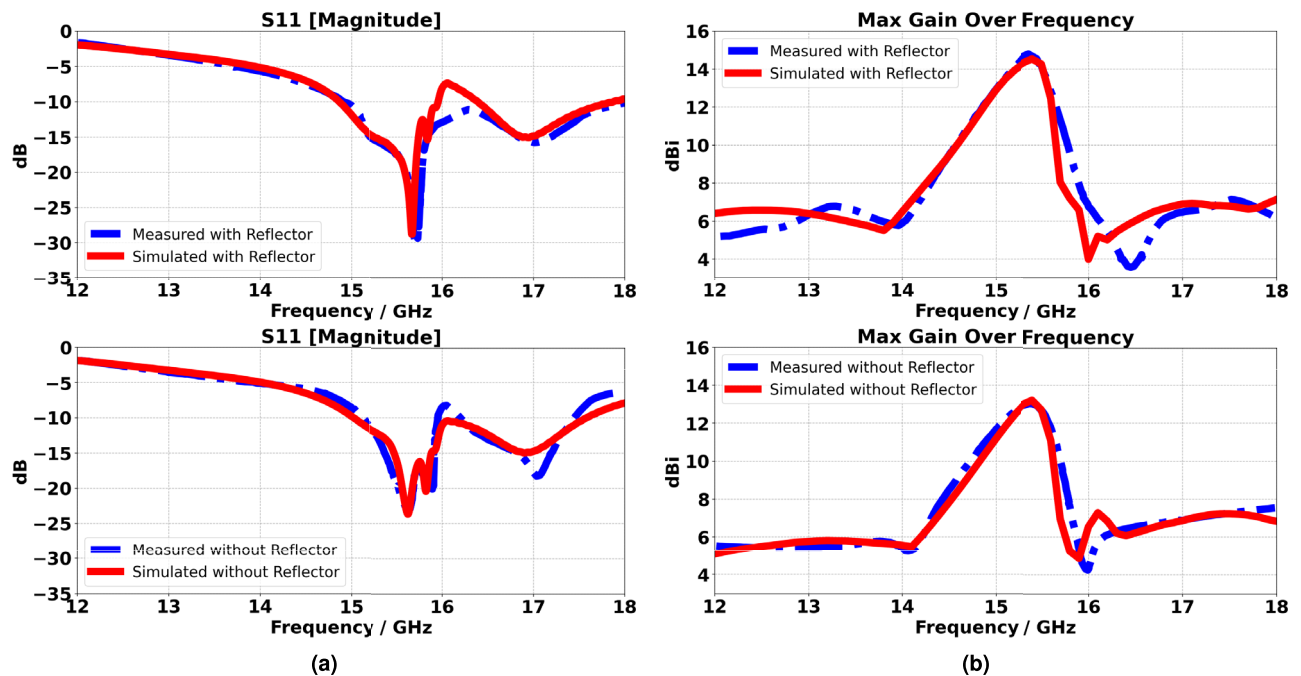


Fig. 21. Comparison between the simulated and fabricated antennas with and without the reflector in terms of: (a) S_{11} , (b) Gain.

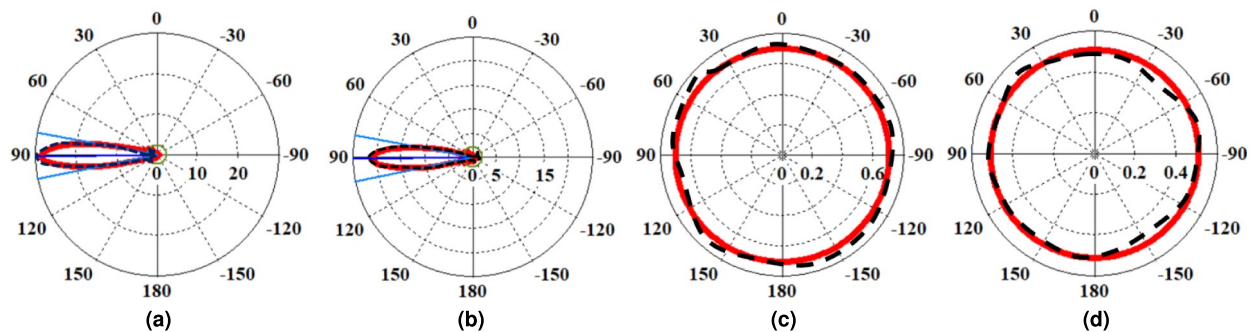


Fig. 22. E-field radiation patterns comparison between the simulated and fabricated antennas with and without the reflector: (a) With reflector co-polarized, (b) Without reflector co-polarized, (c) With reflector cross-polarized, and (d) Without reflector cross-polarized. Note: The solid red line shows the simulated results, while the dashed black line shows the measured data.

and the depth of the S_{11} null. Measurement uncertainty according to fabrication tolerances ($\approx \pm 0.1$ mm), SMA connector loss (≈ 0.2 dB), and anechoic chamber alignment errors ($\approx \pm 2^\circ$) contributed to discrepancies between simulated and measured results. The overall gain measurement uncertainty is estimated at ± 0.5 dBi, and S_{11} accuracy is within ± 1.5 dB across the band. These uncertainties are within acceptable limits for Ku-band antenna characterization. Measurements were performed in a fully anechoic chamber; alternative methods, such as reverberation chambers⁵⁰ could be explored for efficiency characterization in future work. To quantitatively examine fabrication sensitivity, a parametric analysis was performed by varying key parameters within typical PCB manufacturing limits: substrate thickness ± 0.05 mm, trace width ± 0.02 mm, and reflector spacing ± 0.2 mm. Over 100 simulated trials, the resonant frequency shifted by ≤ 0.12 GHz with variations in the BW of 0.04 GHz, gain varied within ± 0.3 dBi, and S_{11} remained below -10 dB across the band in 95% of cases. These results confirm the design's robustness to fabrication uncertainties and align with the observed discrepancies between simulated and measured results, which fall within these tolerance bounds.

Performance comparison

In Table 3, a comparison study is applied between the obtained results and those published in the literature. Various antenna designs based on their size, normalized size relative to the free-space wavelength, resonance frequency, BW, gain, aperture efficiency, substrate, and design methods are listed in this table. The comparison

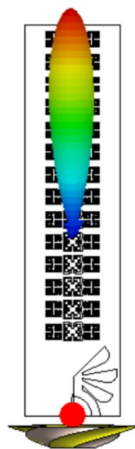


Fig. 23. The proposed antenna 3D radiation pattern at 15.4 GHz.

Ref.	Size (mm ²)	Normalized Size (λ ²)	Resonance Frequency (GHz)	BW (GHz)	Gain (dBi)	Aperture Efficiency (%)	Substrate	Design Method
16	87 × 60	1.8λ × 1.1λ	5.5	2.25–7	2.5	8.1	FR-4, ε _r = 4.4	Two half-Vivaldi + slot
17	187.5 × 190	2.38λ × 2.41λ	3.8	0.56–7.7	9.2	43.5	FR-4, ε _r = 4.4	Dual-polarized Vivaldi pair
18	5.7 × 13.2	0.49λ × 1.13λ	25.85	24.25–27.5	~0.5	12.7	Rogers RO4003C, ε _r = 3.55	Grating Vivaldi
19	25 × 40	0.29λ × 0.46λ	3.5	3.4–3.78	~0	58.4	Rogers RO4003C, ε _r = 3.55	Vivaldi + Marchand balun
20	29 × 33	0.56λ × 0.64λ	5.88	5.825–5.925	4.35	58.8	PTFE, ε _r = 2.1	MIMO patch + defected ground structure
21	29 × 29	0.33λ × 0.33λ	3.5	0.68	0.54	78.7	FR-4, ε _r = 2.1	Asymmetric-slit patch
This work	21.87 × 90.5	1.12λ × 4.64λ	15.4	14.8–16, 16.4–18	14.5	43	Rogers RT5880, ε _r = 2.2	CRLH + 3D reflector

Table 3. Benchmarking performance via comparative analysis.

shows that the proposed antenna realizes the highest gain with a wide BW to establish overall enhancement compared to the published results with a miniaturized size.

Conclusion

This work proposed and experimentally validated a high-gain Vivaldi antenna integrated with a hybrid Hilbert-Minkowski fractal CRLH array of 14-unit cells and a 3D hexagonal reflector for Ku-band communication systems. The simulated results demonstrated that the CRLH array enhanced gain by 7.3 dBi over the baseline design and extended the operating BW through improved phase control. The fabricated antenna prototype achieved a peak gain of 14.5 dBi at 15.4 GHz with an aggregated BW of 2.8 GHz (14.8–16 GHz and 16.4–18 GHz) in close agreement with simulations. The full design, which reduced the SLL and BLL to -10.6 dB and -2.6 dB, respectively, and increased the front-to-back ratio to 17.1 dB, improved the gain by 1.3 dBi and the front-to-back ratio by 3 dB compared to the CRLH-based Vivaldi configuration. The system evaluation using measured radiation characteristics demonstrated a 91.38% reduction in BER and an 11.53% enhancement in CC at -4 dB SNR compared to the CRLH-based Vivaldi configuration under AWGN channel conditions. The ANN-based RLC extraction method was efficiently implemented, yielding precise CRLH dispersion modeling with $R^2 = 0.992$ and $RMSE = 0.032$, outperforming conventional curve-fitting methods in both accuracy and computational efficiency. The suggested design offers a compact, efficient antenna solution for Ku-band applications, such as satellite and V2X communications. Future work will include the design of MIMO configurations and incorporate realistic fading channel models (e.g., 3GPP for V2X, ITU-R for satellite) for comprehensive system-level validation.

Data availability

The datasets generated during the current study are not publicly available due to confidentiality agreements and copyright restrictions enforced by the University of Malaga. However, data may be available from the corresponding author upon reasonable request and with permission from the University of Malaga.

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Author contributions

Mustafa Mahdi Ali applied the theoretical simulations, performed the computations and discussed the results. Enrique Márquez Segura proposed the research problem and supervised the findings of this work. Taha A. Elwi developed the theory and made the fabrication and measurements.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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